PLL
(PHASE LOCKED LOOP)

CLOCK OPTIONS FOR LPC21xx

XTAL1:

Frequency range is 1 MHZ – 50 MHZ

XTAL1-XTAL2:

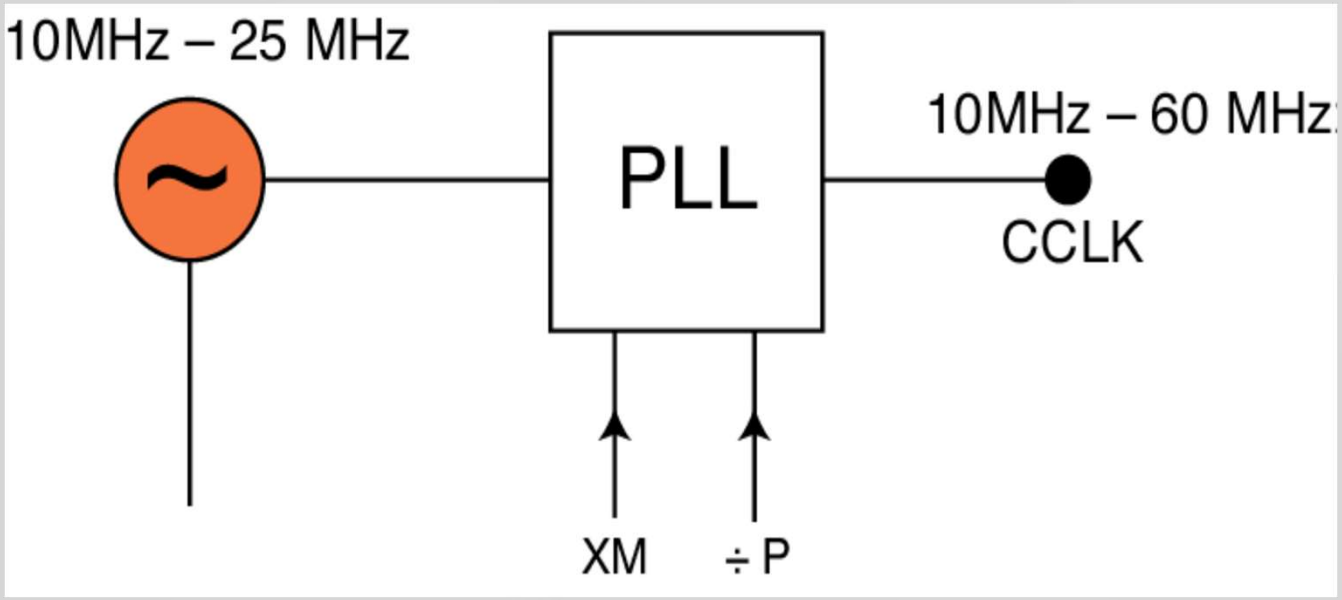
Frequency range is 1 MHZ – 30 MHZ

On-chip PLL with External crystal:

External crystal frequency range is 10 MHZ – 25 MHZ

What is PLL?

PLL stands for Phase-Locked Loop, and it is an electronic feedback control system that is widely used in a variety of applications in electronics and telecommunications.



Why to Use PLL?

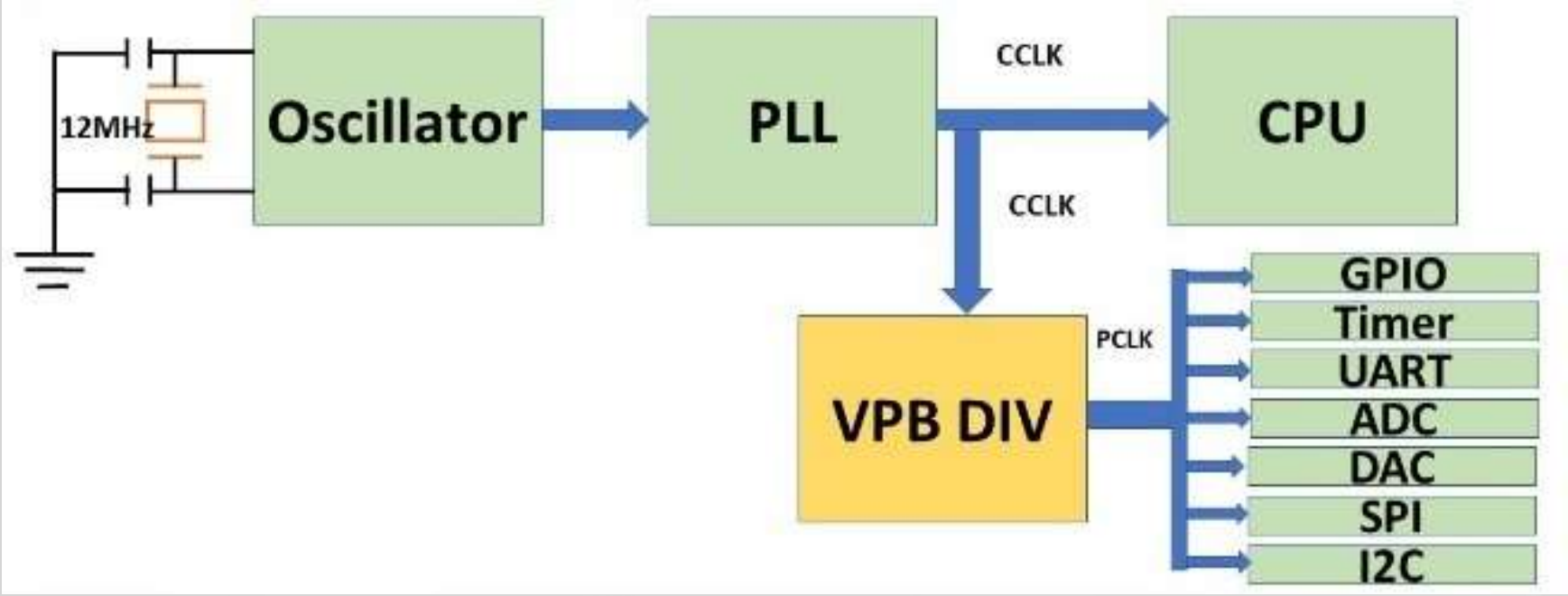
Allows the LPC2000 to run at its maximum frequency with a low value external oscillator, thus minimizing the EMC emissions (i.e. noise) of the LPC2000 series

LPC2129:
Only one PLL module is available in LPC2129.

LPC2148:
There are two PLL modules in the LPC2141/2/4/6/8 microcontroller. The PLL0 is used to generate the CCLK clock (system clock) while the PLL1 has to supply the clock for the USB at the fixed rate of 48 MHz.

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PHASE LOCKED LOOP



PHASE LOCKED LOOP

The PLL accepts an input clock frequency in the range of 10 MHz to 25 MHz only.

The input frequency is multiplied up into the cclk with the range of 10 MHz to 60 MHz using a Current Controlled Oscillator (CCO).

The multiplier can be an integer value from 1 to 32 (in practice, the multiplier value cannot be higher than 6 on the LPC2119/2129/2194/2292/2294 due to the upper frequency limit of the CPU).

The CCO operates in the range of 156 MHz to 320 MHz, so there is an additional divider in the loop to keep the CCO within its frequency range while the PLL is providing the desired output frequency.

The output divider may be set to divide by 2, 4, 8, or 16 to produce the output clock.

Address	Name	Description	Access
0xE01FC080	PLLCON	PLL Control Register. Holding register for updating PLL control bits. Values written to this register do not take effect until a valid PLL feed sequence has taken place.	R/W
0xE01FC084	PLLCFG	PLL Configuration Register. Holding register for updating PLL configuration values. Values written to this register do not take effect until a valid PLL feed sequence has taken place.	R/W
0xE01FC088	PLLSTAT	PLL Status Register. Read-back register for PLL control and configuration information. If PLLCON or PLLCFG have been written to, but a PLL feed sequence has not yet occurred, they will not reflect the current PLL state. Reading this register provides the actual values controlling the PLL, as well as the status of the PLL.	RO
0xE01FC08C	PLLFEED	PLL Feed Register. This register enables loading of the PLL control and configuration information from the PLLCON and PLLCFG registers into the shadow registers that actually affect PLL operation.	WO

Note: To know in detailed information about each register, Refer LPC2129 data sheet

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Basic Terms/symbols

FOSC	=> frequency from the crystal oscillator(XTAL)/external clock
FCCO	=> frequency of the PLL Current Controlled Oscillator(CCO)
CCLK	=> PLL output frequency (CPU Clock)
M	=> PLL Multiplier value from the MSEL bits in the PLLCFG register
P	=> PLL Divider value from the PSEL bits in the PLLCFG register
PCLK	=> Peripheral Clock which is derived from CCLK

Calculations

PLL output clock is

CCLK = M x FOSC

=or=

CCLK = FCCO / (2 x P)

CCO output clock is

FCCO = CCLK x 2 x P

=or=

FCCO = FOSC x M x 2 x P

Note: FCCO range is 156 MHz to 320 MHz

For 156MHz, P = 1.3
For 320MHz, P = 2.67, so we will use P=2

The PLL inputs and settings must meet the following:

10 MHz>=FOSC<=25 MHz.

10MHz>=CCLK<=Fmax (depends on the CPU selection)

156 MHz>=FCC0<=320 MHz.

HOW TO SET UP PLL?

The **Order** of setting up PLL is **strictly** as follows :
Setup PLL
Apply Feed Sequence
Wait for PLL to lock
Then Connect PLL
Apply Feed Sequence

```
#include <LPC21xx.H>

#define PLOCK 0x00000400

void setupPLL0(void);
void feedSeq(void);
void connectPLL0(void);

int main(void)
{
   setupPLL0();
   feedSeq(); //sequence for locking PLL to desired freq.
   connectPLL0();
   feedSeq(); //sequence for connecting the PLL as system clock
   //SysClock is now ticking @ 60Mhz!
   VPBDIV = 0x01; // PCLK is same as CCLK i.e 60Mhz
   while(1);
   //return 0;
}
```

```
void setupPLL0(void)
{
   PLLCON = 0x01; // PLLE=1 & PPLC=0 so it will be enabled
                  // but not connected after FEED sequence
   PLLCFG = 0x24; // set the multipler to 5 (i.e actually 4)
                  // i.e 12x5 = 60 Mhz (M - 1 = 4)!!!
                  // Set P=2 since we want FCCO in range!!!
                  // So , Assign PSEL =01 in PLL0CFG as per the table.
}
void feedSeq(void)
{
   PLLFEED = 0xAA;
   PLLFEED = 0x55;
}
```

```
void connectPLL0(void)
{
   // check whether PLL has locked on to the  desired freq by reading the lock bit in the PPL0STAT register
   while( !( PLLSTAT & PLOCK ));
   // now enable(again) and connect
   PLLCON = 0x03;
}
```